

Conditional Statements Assignment

Question 1: Student Result Checker

A school wants to automatically determine whether a student has passed or failed.

Ask the user to enter the student's marks.

- If marks are 50 or above, display "Pass".
- Otherwise, display "Fail".

Write a Python program to implement this system.

Question 2: Electricity Bill Category

An electricity company wants to categorize customers based on their monthly electricity units.

Ask the user to enter the number of units consumed.

- Less than 100 units → "Low Usage"
- 100 to 300 units → "Medium Usage"
- More than 300 units → "High Usage"

Display the appropriate category.

Question 3: Online Shopping Discount

An online store gives discounts based on the total purchase amount.

Ask the customer for the total amount.

- Less than 5,000 → No discount
- 5,000 to 9,999 → 10% discount
- 10,000 to 19,999 → 15% discount
- 20,000 or above → 20% discount

Calculate and display:

- Original amount
 - Discount
 - Final amount
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Question 4: Employee Salary Grade

A company wants to assign salary grades based on an employee's salary.

Ask the employee to enter their salary.

- Below 30,000 → Grade C
- 30,000 to 59,999 → Grade B
- 60,000 to 99,999 → Grade A
- 100,000 or above → Grade A+

Display the employee's salary grade.

Question 5: Login System

You are developing a simple login system.

The correct username is:

admin

The correct password is:

python123

Ask the user to enter their username and password.

- If both are correct, display "Login Successful".
 - If the username is incorrect, display "Invalid Username".
 - If the username is correct but the password is incorrect, display "Invalid Password".
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Question 6: Movie Ticket Pricing

A cinema charges different ticket prices based on age.

Ask the customer to enter their age.

- Under 5 → Free
- 5 to 12 → Rs. 300
- 13 to 59 → Rs. 600
- 60 or above → Rs. 400

Display the ticket price.

Question 7: Number Analyzer

A data-entry application needs to analyze a number entered by the user.

Ask the user to enter a number.

Your program should determine:

- Whether the number is positive, negative, or zero.
- If it is positive or negative, also determine whether it is even or odd.

For example:

Enter number: -7

Negative

Odd

Question 8: ATM Withdrawal

You are developing an ATM withdrawal system.

The user's account balance is:

50000

Ask the user to enter the withdrawal amount.

Apply these rules:

- If the amount is less than or equal to 0, display "Invalid Amount".
 - If the amount is greater than the balance, display "Insufficient Balance".
 - Otherwise, process the withdrawal and display the remaining balance.
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Question 9: University Admission Eligibility

A university wants to automatically check whether a student is eligible for admission.

Ask the student for:

- Matric percentage
- Intermediate percentage

Eligibility requirements:

- Matric percentage must be at least 60.
- Intermediate percentage must be at least 60.

If both conditions are satisfied:

Eligible for Admission

Otherwise:

Not Eligible

Question 10: Job Eligibility System

A company wants to filter candidates for a job.

Ask the candidate for:

- Age
- Years of experience
- Technical test score

Eligibility rules:

- Age must be at least 18.
- Experience must be at least 1 year.
- Technical test score must be at least 60.

If all requirements are satisfied, display:

Candidate is Eligible

Otherwise, display the specific reason why the candidate is not eligible

Question 11: Smart Student Admission System

A university wants to create a student admission screening system.

Ask the student to enter:

- Student name
- Age
- Matric percentage
- Intermediate percentage
- Entry test score
- Whether the student has a domicile (yes/no)

The university has the following admission rules:

Basic Eligibility

The student must:

- Be at least 17 years old.

- Have at least 60% in Matric.
- Have at least 60% in Intermediate.
- Have at least 50% in the entry test.
- Have a valid domicile.

If the student fails any requirement, display **Not Eligible** and clearly mention the reason.

Merit Category

If the student is eligible, determine their merit category:

Entry Test ≥ 85 AND Intermediate ≥ 85

→ Excellent Merit

Entry Test ≥ 70 AND Intermediate ≥ 70

→ Good Merit

Entry Test ≥ 50 AND Intermediate ≥ 60

→ Normal Merit

Otherwise:

Not Eligible

Scholarship

For eligible students:

Merit $\geq$ Excellent Merit

→ 50% Scholarship

Merit == Good Merit

→ 25% Scholarship

Otherwise

→ No Scholarship

Final Display

===== ADMISSION REPORT =====

Name: Ali

Age: 19

Intermediate: 82%

Entry Test: 88%

Status: Eligible

Merit: Excellent Merit

Scholarship: 50%

Question 12: Employee Performance & Promotion System

A company wants to automatically evaluate its employees for promotion.

Ask the employee for:

- Employee name
- Years of experience
- Monthly salary
- Performance score out of 100
- Number of projects completed
- Number of late days

Promotion Rules

An employee can be promoted only if:

- Experience is at least 2 years.
- Performance score is at least 70.
- At least 3 projects have been completed.
- Late days are less than 10.

If any requirement is not satisfied, the employee is **Not Eligible for Promotion**.

The program should display the specific reason.

Promotion Level

If the employee is eligible:

Performance ≥ 90

AND Experience ≥ 5

→ Senior Promotion

Performance ≥ 80

AND Experience ≥ 3

→ Mid-Level Promotion

Otherwise

→ Junior Promotion

Bonus

Calculate the bonus category:

Performance >= 90

→ 20% bonus

Performance >= 80

→ 15% bonus

Performance >= 70

→ 10% bonus

Otherwise

→ No bonus

Finally, display:

===== EMPLOYEE REPORT =====

Employee: Ahmed

Experience: 4 years

Performance: 87

Projects: 6

Late Days: 3

Promotion Status: Eligible

Promotion Level: Mid-Level

Bonus: 15%

Concepts Covered in this Assignment

- Variables
- Input
- Data types
- Comparison operators
- Logical operators
- if
- elif
- else
- Nested conditions
- Strings
- f-strings